



Air Quality Index Prediction



Forecasting pollution levels based on environmental data and AI analytics.

Action Project Reference Number: 57-15

Action Project Title: Air Quality Index Prediction – Forecasting pollution levels based on environmental data and AI analytics

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TERMS OF REFERENCE

This report has been prepared for Solar Fiji as part of Think Pacific Action Project 57-15.

The project brief focused on using environmental data and artificial intelligence to forecast air pollution and Air Quality Index conditions. The wider purpose was to explore whether this type of forecasting could help Solar Fiji better understand environmental conditions that may be relevant to solar-energy production, maintenance and operational planning.

The project was completed using existing environmental datasets, historical air-quality information, weather data, statistical forecasting and machine-learning techniques. Different forecasting approaches were compared before a final prototype was developed.

The finished project is an environmental forecasting and decision-support dashboard. It combines a customised PM2.5 prediction, AQI interpretation, weather conditions, solar-radiation information, forecast uncertainty, an external CAMS comparison and a local site-condition reporting feature.

The project should be viewed as a proof of concept rather than a finished commercial system. Its purpose is to demonstrate what could be achieved, identify the limitations that would need to be addressed and suggest a realistic pathway for future development.

EXECUTIVE SUMMARY

Solar Fiji operates in an environment where geography, weather and access to remote locations can all influence the way solar systems are monitored and maintained. The original Action Project asked whether environmental information and artificial intelligence could be used to forecast air-quality conditions and support Solar Fiji in anticipating changing environmental conditions.

To explore this challenge, a pilot forecasting system was developed for Suva and Nasinu. Historical PM2.5 air-quality data were combined with environmental information including temperature, humidity, rainfall, wind speed, wind direction, atmospheric pressure and solar radiation. A live-compatible forecasting pipeline was then created so that recent environmental information and forecast weather could be used to produce current predictions.

Three main forecasting approaches were evaluated: a persistence baseline, ARIMAX and XGBoost. The persistence model assumed that future PM2.5 conditions would remain similar to the previous day. ARIMAX combined historical air-quality patterns with environmental variables, while XGBoost provided a machine-learning alternative.

ARIMAX(1,0,1) achieved the strongest overall performance on the held-out test period. It produced a Mean Absolute Error of 1.368 $\mu\text{g}/\text{m}^3$, a Root Mean Squared Error of 1.745 $\mu\text{g}/\text{m}^3$ and an R^2 value of 0.390. This represented a 17.25% improvement in RMSE compared with the persistence baseline. XGBoost also performed strongly, achieving an RMSE of 1.776 $\mu\text{g}/\text{m}^3$.

An important limitation was also identified. Most historical days fell within the Good AQI category, meaning there were relatively few Moderate air-quality events for the models to learn from. Although ARIMAX produced the lowest overall forecasting error, it did not successfully identify the small number of Moderate AQI days in the final test period.

The final dashboard was therefore designed not to rely on a single model output alone. It also displays a forecast confidence range, an external CAMS comparison, expected weather conditions and a local site-reporting feature.

The project considered three possible implementation approaches: using external forecasts alone, using the hybrid decision-support system developed during this project, or creating a future sensor-integrated platform. The hybrid approach is recommended because it provides a useful balance between cost, practicality, customisation and future scalability.

The recommended next step is to pilot the prototype in Suva and Nasinu, gather feedback from Solar Fiji staff and begin collecting site-level generation, maintenance and environmental observations. These data would make it possible to evaluate the relationship between environmental conditions and actual solar-system performance more directly.

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1. INTRODUCTION & PROJECT CONTEXT

1.1 Project Background

This Action Project focused on the use of artificial intelligence and environmental data to forecast Air Quality Index conditions.

The original brief highlighted the potential relationship between poor air quality and solar operations. Conditions such as smoke, dust and airborne particles may affect the amount of solar radiation reaching photovoltaic panels or contribute to deposits on panel surfaces.

The project therefore explored whether air-quality forecasting could provide Solar Fiji with an additional source of information when monitoring changing environmental conditions.

The aim was not simply to build an AI model. The wider goal was to turn environmental data into something practical and understandable that could potentially support real operational decisions.

1.2 The Organisational Context

Solar Fiji works in renewable energy across a geographically dispersed island environment.

Solar installations may be located in very different settings, including:

- urban areas;
- commercial locations;
- rural communities;
- coastal areas;
- remote islands.

These locations can experience different weather and environmental conditions.

Access to remote sites may also require additional time, staff and travel. This means that useful forward-looking environmental information could potentially help teams decide when a site may require closer monitoring.

Environmental forecasts already exist, but they are usually designed for general atmospheric or weather monitoring.

They are not normally presented specifically around questions such as:

- What environmental conditions are expected?
- How confident is the forecast?
- Is anything unusual developing?

- Should a solar site be monitored more closely?
- Have local technicians observed something that regional models may have missed?

This project explored how those questions could be brought together in one simple system.

1.3 Research Question

The main research question was:

How can environmental data and AI-based forecasting be combined to predict short-term air-quality conditions and provide practical decision support for Solar Fiji's solar operations?

1.4 Aim

The aim of the project was to design and evaluate an environmental forecasting prototype that combines air-quality prediction, weather information, machine learning and practical operational guidance for Solar Fiji.

1.5 Objectives

The project aimed to:

1. Understand the environmental and organisational context behind the Action Project.
2. Identify appropriate historical and current environmental data sources.
3. Analyse historical PM2.5 and weather patterns.
4. Develop and compare forecasting models.
5. Build a live-compatible forecasting pipeline.
6. Translate model outputs into a simple decision-support dashboard.
7. Identify practical barriers to implementation.
8. Explore different implementation options.
9. Recommend a realistic future development pathway.

1.6 Project Scope

The initial pilot focused on Suva and Nasinu.

The primary forecast target was daily PM2.5 concentration. The predicted PM2.5 value was then converted into an AQI interpretation for the dashboard.

The most rigorously evaluated forecasting horizon was one day ahead.

A seven-day environmental outlook was also included in the prototype, although predictions beyond the primary horizon should be treated as broader planning information with greater uncertainty.

The project did not directly model solar electricity generation because site-level Solar Fiji generation data were not available.

Traffic and industrial activity were considered during the project design process because they were mentioned in the original brief. However, suitable consistent local datasets were not identified within the available timeframe.

Rather than introducing weak or unreliable proxy variables, these factors were left for future development.

2. FIJI, SUSTAINABILITY AND PROJECT ALIGNMENT

2.1 Fiji and the Environmental Context

Fiji's geography is an important consideration for any technology-based proposal.

Solar systems may operate in urban, rural, coastal and remote island environments.

Conditions experienced at one location may therefore be very different from those experienced at another.

Any forecasting system designed for Solar Fiji should therefore be flexible enough to develop over time rather than assuming that a single location represents every site.

For this reason, the current system begins with a Suva and Nasinu pilot rather than attempting to immediately create a model for every Solar Fiji installation.

2.2 Environmental Sustainability

Solar Fiji's work contributes directly to renewable-energy development.

This project supports that wider environmental objective by exploring how environmental information can be used alongside solar operations.

The prototype brings together information on:

- air quality;
- weather;
- rainfall;
- wind;
- solar radiation;
- seasonal conditions.

This creates a broader picture of the environment surrounding solar operations.

2.3 Financial Considerations

Better environmental information could potentially support more informed maintenance planning.

For example, if future research shows that certain environmental conditions are regularly followed by panel-soiling or performance issues, Solar Fiji could use that information to prioritise inspections.

However, this project did not directly measure financial savings.

It would therefore be inappropriate to claim that the prototype currently reduces maintenance costs.

Actual cost benefits would need to be evaluated through a longer operational pilot.

2.4 Social and Communication Considerations

Environmental data can quickly become difficult to understand.

Values such as PM2.5 concentration, statistical confidence intervals and forecasting errors may be useful to technical users but less useful to someone who simply needs to know whether conditions require attention.

For this reason, the prototype uses simple descriptions such as:

- Good air quality;
- Moderate air quality;
- Solar operating conditions;
- Continue monitoring;
- Check for haze, smoke or visible panel deposits if performance falls.

The more detailed technical information remains available for users who need it.

2.5 Respecting Local Knowledge

One of the most important design principles was that the forecasting system should support local experience rather than replace it.

A global atmospheric model may indicate relatively normal conditions while a technician at a specific site is observing:

- smoke;
- local dust;
- salt deposits;
- dirty panels;
- unusual weather;
- reduced solar output.

For this reason, the dashboard includes a site-condition reporting feature.

Users can record what they are seeing at a particular location and whether they have noticed unusual solar performance.

This creates the possibility of combining model-based forecasting with real local observations in the future.

3. RESEARCH METHOD AND DATA

3.1 Research Approach

The project mainly used quantitative analysis based on existing environmental data.

The overall process involved:

- collecting existing datasets;
- cleaning and combining data;
- analysing historical patterns;
- creating forecasting variables;
- developing models;
- comparing performance;
- integrating current environmental information;
- developing the dashboard.

The approach was iterative.

Some methods were adjusted as limitations became clearer during the project.

This was particularly important when developing the live forecasting system, where historical and current data needed to be prepared in a compatible format.

3.2 Pilot Location

The pilot focused on Suva and Nasinu.

The approximate coordinates used were:

Latitude: -18.1416

Longitude: 178.4419

This area provided a practical starting point for developing and testing the prototype.

3.3 Historical Air-Quality Data

Historical PM2.5 information was obtained from an existing CAMS-derived dataset.

The broader historical dataset covered approximately: **1 January 2021 to 10 July 2024**

For the final live-compatible modelling pipeline, the main period was: **1 January 2022 to 10 July 2024**

After creating historical lag variables, the final modelling dataset covered:

15 January 2022 to 10 July 2024

The final model dataset contained: **908 daily observations**

3.4 Environmental Data

Weather information included:

- mean temperature;
- maximum temperature;
- minimum temperature;
- relative humidity;
- rainfall;
- wind speed;
- wind direction;
- surface pressure;
- shortwave solar radiation.

Seasonal and calendar variables were also included.

Historical and current weather information were prepared using a consistent GFS-based workflow.

3.5 Historical PM2.5 Features

The machine-learning model also used historical PM2.5 information.

Examples included:

- previous-day PM2.5;
- two-day lag;
- three-day lag;
- seven-day lag;
- fourteen-day lag;
- short-term rolling averages;
- longer-term rolling averages;
- recent changes;

- recent pollution trends.

All historical features were created using previous information only.

This was important to avoid data leakage.

3.6 Data Quality

The dataset was checked for:

- missing values;
- duplicate dates;
- missing calendar days;
- invalid values;
- inconsistent units;
- chronological errors.

The final modelling dataset did not contain missing values in the selected model features.

3.7 Train, Validation and Test Periods

The final data were divided chronologically.

Training period

15 January 2022 to 11 October 2023

635 observations

Validation period

12 October 2023 to 24 February 2024

136 observations

Test period

25 February 2024 to 10 July 2024

137 observations

A random split was not used because forecasting should be evaluated on future information.

The models were trained on earlier periods and then tested on dates they had not previously seen.

4. DATA ANALYSIS AND KEY FINDINGS

4.1 Historical Air Quality

Across the broader historical dataset, average PM_{2.5} concentration was approximately: **5.82 µg/m³**

The median concentration was approximately: **5.50 $\mu\text{g}/\text{m}^3$**

The highest daily value was approximately: **20.24 $\mu\text{g}/\text{m}^3$**

Most days were classified as Good.

The approximate historical AQI distribution was:

Good: 89.28%

Moderate: 10.72%

No higher AQI categories were represented in the available historical data.

This became an important consideration when interpreting model performance. Because Good days were much more common, a model could achieve high overall AQI accuracy while still performing poorly on the less common Moderate days.

4.2 Seasonal Patterns

The average historical PM_{2.5} concentration was higher during the dry season.

Dry season: approximately 6.28 $\mu\text{g}/\text{m}^3$

Wet season: approximately 5.38 $\mu\text{g}/\text{m}^3$

This suggests that seasonal conditions may influence pollution patterns.

However, the data alone do not identify the exact cause of the seasonal difference.

4.3 Environmental Relationships

Several environmental variables showed relationships with PM_{2.5}.

Wind speed had one of the strongest simple relationships among the weather variables analysed.

Temperature, humidity and solar-radiation variables also showed associations.

These findings supported the decision to include environmental conditions in the forecasting models.

However, these relationships should not automatically be interpreted as cause and effect.

The analysis shows association, not proof of causation.

4.4 What the Historical Analysis Means

The historical analysis suggested three main points.

First, pollution levels in the available dataset were generally low.

Second, short-term PM2.5 behaviour appeared to contain useful information for predicting future conditions.

Third, environmental variables added useful context and potential predictive information.

These findings guided the final model-development stage.

5. FORECASTING MODELS AND EVALUATION

5.1 Persistence Baseline

A persistence model was used as the basic benchmark.

This assumes that the next day's PM2.5 level will be similar to the previous day's value.

The purpose of the baseline was to answer a simple question:

Does the more advanced forecasting system actually perform better than just using yesterday's value?

5.2 ARIMAX

ARIMAX combines historical time-series behaviour with additional environmental variables.

Several ARIMAX configurations were tested during validation.

The strongest configuration was: **ARIMAX(1,0,1)**

The model used environmental inputs including:

- temperature;
- humidity;
- rainfall;
- wind;
- surface pressure;
- solar radiation;
- seasonal information.

ARIMAX was selected as the primary production model because it achieved the strongest overall test performance.

5.3 XGBoost

XGBoost was developed as an alternative machine-learning model.

It used a wider set of features, including historical PM2.5 lags and environmental information.

Its final configuration used a relatively shallow tree structure with regularisation to reduce the risk of overfitting.

XGBoost was retained in the final system even though ARIMAX performed slightly better.

It provided two useful benefits:

- 1. A strong comparison against the selected model.
- 2. A way to examine which variables contributed most strongly to prediction.

5.4 LSTM Exploration

LSTM was also explored during the broader modelling process because it was recommended in the original Action Project brief.

However, it did not outperform the simpler approaches.

The available dataset was also relatively small compared with the amount of data often used for deep-learning time-series systems.

For this reason, LSTM was not retained in the final live-compatible system.

This was still a useful result.

The project showed that a more complex AI model is not automatically the best option.

5.5 Final Model Performance

The final test results were:

Model	MAE	RMSE	R ²	sMAPE
ARIMAX(1,0,1)	1.368	1.745	0.390	27.40%
XGBoost	1.411	1.776	0.368	28.45%
Persistence	1.540	2.108	0.109	31.33%

ARIMAX achieved the lowest overall prediction error.

Compared with persistence, its RMSE improved by approximately: **17.25%**

XGBoost improved on persistence by approximately: **15.79%**

The difference between ARIMAX and XGBoost was relatively small.

This suggests that both approaches captured useful information from the data.

5.6 AQI Category Performance

The AQI evaluation revealed an important limitation.

Model	AQI Accuracy	Macro F1	Weighted F1	Moderate Recall
Persistence	0.934	0.683	0.932	0.375
ARIMAX	0.942	0.485	0.913	0.000
XGBoost	0.942	0.651	0.932	0.250

ARIMAX achieved approximately 94.2% overall AQI-category accuracy.

However, it did not correctly identify any of the eight Moderate days in the final test period.

The main reason this matters is that overall accuracy can appear strong when one category is much more common than another.

The test period contained: **129 Good days**

and only: **8 Moderate days**

The dashboard therefore includes additional information instead of relying entirely on the ARIMAX category prediction.

This includes:

- forecast uncertainty;
- external CAMS comparison;
- environmental conditions;
- local observations.

5.7 What Influenced the Forecast?

XGBoost feature importance was used to understand which variables contributed most strongly to prediction.

The most influential variables included:

1. Previous-day PM2.5.
2. Recent three-day PM2.5 average.
3. Wind speed.
4. Recent PM2.5 change.
5. PM2.5 from two days earlier.
6. Maximum temperature.
7. Mean temperature.

8. Fourteen-day PM2.5 trend.
9. Rainfall.
10. Solar radiation.

The results suggest that recent pollution behaviour was particularly important.

Environmental conditions also contributed to the forecast.

These results show predictive importance rather than proven physical causation.

6. DISCUSSION, CHALLENGES AND LIMITATIONS

6.1 What the Results Mean for Solar Fiji

The project demonstrates that environmental data can be transformed into useful forward-looking information.

The main value of the project is not the prediction model by itself.

The stronger contribution is the combination of:

- customised forecasting;
- external environmental information;
- uncertainty;
- weather conditions;
- understandable interpretation;
- local observations.

Together, these elements create a more practical decision-support system.

6.2 Clear Findings

Several findings can be stated with reasonable confidence.

First, ARIMAX and XGBoost both outperformed the simple persistence baseline.

Second, ARIMAX achieved the strongest overall PM2.5 prediction performance.

Third, recent PM2.5 levels were important predictors of future conditions.

Fourth, environmental information such as wind and temperature contributed to prediction.

Fifth, current environmental feeds can be integrated into a working forecasting dashboard.

6.3 Findings That Need Further Research

Some questions cannot be answered confidently from the current project.

The project does not prove how much air pollution affects actual Solar Fiji electricity generation.

This would require measured generation data.

The project also cannot confidently demonstrate model performance during severe pollution events because these conditions were not represented in the historical dataset.

Further research would also be needed to understand whether the Suva and Nasinu model would perform equally well across:

- remote islands;
- coastal sites;
- agricultural areas;
- locations with very different pollution sources.

6.4 Financial Challenges

The current prototype uses mainly accessible data and open-source software.

However, a future operational system could involve costs for:

- permanent hosting;
- sensors;
- system maintenance;
- technical support;
- premium environmental data.

A staged approach would therefore be more realistic than immediately investing in a complex monitoring system.

6.5 Resource Challenges

A production forecasting platform would require ongoing maintenance.

This could include:

- maintaining data connections;
- checking model performance;
- retraining models;
- updating software;
- managing failures.

The long-term system would therefore require technical ownership.

6.6 Communication Challenges

A useful operational tool must be understandable.

Technical outputs such as confidence intervals and PM2.5 concentrations should therefore be accompanied by plain-language interpretation.

The dashboard was designed with this in mind.

Future versions could go further by providing:

- mobile notifications;
- downloadable summaries;
- low-bandwidth pages;
- simple daily reports.

6.7 Data Limitations

The most significant limitations were related to data.

Historical PM2.5 and current operational PM2.5 information are not produced through exactly the same process.

The historical dataset also contained very few Moderate AQI events and no poor-air-quality categories beyond Moderate.

No local ground-monitoring station was available for independent validation.

Solar Fiji generation data were not available.

Traffic and industrial-activity information were not included.

These limitations mean that the prototype should be treated as an environmental decision-support tool rather than an automated operational system.

7. OPTIONS EXPLORED AND RECOMMENDED PROPOSAL

Three possible approaches were considered.

7.1 Option 1 – External Forecast Dashboard

The first option would simply display existing weather and air-quality forecasts.

Advantages

- low cost;
- easy to implement;
- low technical maintenance.

Disadvantages

- limited customisation;
- no organisation-specific forecasting model;
- less opportunity to learn from historical data;
- limited local input.

7.2 Option 2 – Hybrid Forecasting and Decision-Support System

This is the approach developed during the project.

It combines:

- ARIMAX prediction;
- external CAMS comparison;
- weather information;
- forecast uncertainty;
- XGBoost explanation;
- local site observations.

Advantages

- more customised;
- still relatively low cost;
- provides multiple sources of information;
- supports local knowledge;
- can be expanded gradually.

Disadvantages

- requires technical maintenance;
- still depends on external data;
- requires further validation.

7.3 Option 3 – Sensor-Integrated System

A future system could combine:

- local air-quality sensors;
- solar inverter data;
- local weather information;
- external forecasts;
- machine learning.

Advantages

- greater local relevance;
- direct comparison between environmental conditions and energy generation;
- potential for site-specific models.

Disadvantages

- higher cost;
- sensor maintenance;
- calibration requirements;
- greater technical complexity;
- connectivity requirements.

7.4 Comparison

Option	Cost	Maintenance	Local Relevance	Current Feasibility
External Forecast Only	Low	Low	Moderate	High
Hybrid Prototype	Low–Moderate	Moderate	High	High
Sensor-Integrated Platform	High	High	Very High	Future

7.5 Recommended Proposal

The hybrid forecasting and decision-support approach is recommended.

It offers a practical middle ground.

It provides more value than simply displaying external data but does not immediately require the cost of installing a full monitoring network.

It also allows the system to grow over time.

Future Solar Fiji data could eventually be added without needing to completely redesign the platform.

This makes the hybrid approach the most realistic current option.

8. FINAL PROTOTYPE

The final project output is the Solar Fiji Environmental Forecast Prototype.

The dashboard contains several sections.

8.1 Latest Forecast

The main page shows:

- air-quality category;
- predicted PM2.5;
- expected sunlight;
- solar operating conditions;
- recommended action.

The most important information is shown first. Technical detail remains available without overwhelming the user.

8.2 Environmental Conditions

The forecast also includes:

- temperature;
- rainfall;
- wind speed;
- external CAMS PM2.5 comparison.

This provides more context around the main prediction.

8.3 Forecast Confidence

A 95% forecast interval is included.

This helps communicate that forecasts are not exact.

The dashboard also warns users when the confidence range crosses an AQI-category boundary.

8.4 Seven-Day Outlook

The system displays a seven-day environmental outlook.

The first day represents the primary model horizon.

The following days are included as broader planning guidance and should be interpreted with greater caution.

8.5 Historical Performance

The dashboard includes a historical comparison of:

- ARIMAX;
- XGBoost;
- persistence.

This allows users to see how the selected model performed during testing.

8.6 Forecast Drivers

The feature-importance page shows the main variables that influenced the XGBoost model.

This makes the machine-learning component more understandable.

8.7 Site Condition Reporting

The dashboard also allows users to record local site conditions.

Examples include:

- haze;
- smoke;
- dust;
- salt deposits;
- dirty panels;
- heavy rain;
- strong wind;
- cloud cover;
- reduced solar generation.

This feature is particularly important because local environmental events may not always be fully represented in global forecasting models.

9. RECOMMENDATIONS AND IMPLEMENTATION ROADMAP

9.1 Immediate Stage – 0 to 3 Months

Use the prototype as a small pilot.

Recommended actions include:

- review the dashboard with Solar Fiji staff;
- identify which information is most useful;
- collect usability feedback;
- begin recording local site conditions.

The prototype should remain a decision-support tool during this stage.

9.2 Short Term – 3 to 12 Months

Begin collecting more operational data.

Useful information could include:

- daily energy generation;
- inverter performance;
- cleaning dates;
- maintenance events;
- panel-soiling observations;
- unusual environmental conditions.

This would allow future analysis to examine whether forecast environmental conditions are actually associated with measurable changes in system performance.

9.3 Medium Term – 6 to 18 Months

Test the system across different types of locations.

A useful pilot could include:

- one urban site;
- one coastal site;
- one remote island site.

This would help determine whether different sites require different forecasting approaches.

9.4 Long Term – 12 Months and Beyond

If the prototype demonstrates value, Solar Fiji could consider:

- local air-quality sensors;
- inverter-data integration;
- site-specific models;
- automatic notifications;
- maintenance-priority indicators.

Automated operational decisions should only be introduced after the system has been sufficiently validated with local data.

9.5 Main Recommendations

The main recommendations from the project are:

Recommendation 1: Begin with a limited Suva and Nasinu pilot.

Recommendation 2: Collect site-level generation and maintenance data.

Recommendation 3: Continue to use forecasts as decision support rather than automatic maintenance instructions.

Recommendation 4: Retain local technician observations as part of the system.

Recommendation 5: Re-evaluate model performance as more data become available.

Recommendation 6: Consider sensor integration only after the value of the initial pilot has been demonstrated.

10. CONCLUSION

This Action Project explored how environmental data and AI-based forecasting could support Solar Fiji in understanding short-term air-quality and weather conditions.

The project began with the challenge of forecasting AQI but developed into a wider environmental decision-support prototype.

Historical PM_{2.5} and weather data were analysed, different forecasting approaches were compared and a live-compatible forecasting pipeline was developed.

ARIMAX(1,0,1) achieved the strongest overall performance, with an RMSE of 1.745 $\mu\text{g}/\text{m}^3$ and a 17.25% improvement compared with the persistence baseline.

XGBoost achieved similar results and provided additional value by helping explain which variables were contributing most strongly to predictions.

The project also identified important limitations.

Most historical days had Good air quality, meaning the models had limited opportunities to learn from higher pollution conditions.

The selected ARIMAX model also failed to identify the small number of Moderate AQI days in the test period.

Solar Fiji generation data were not available, meaning the project cannot claim to directly predict solar-efficiency losses.

For these reasons, the prototype should be viewed as a decision-support system rather than an automated decision-making system.

The recommended approach is to begin with a limited pilot, gather local feedback and start collecting operational data.

Over time, the system could be expanded using actual generation information, maintenance records and possibly local environmental sensors.

Overall, the project shows that environmental forecasting, machine learning and local observations can be combined into a practical system designed around Solar Fiji's needs.

11. ACKNOWLEDGEMENTS

I would like to thank Solar Fiji and Think Pacific for the opportunity to work on this Action Project.

The project provided an opportunity to explore how environmental data, artificial intelligence and renewable-energy operations could be brought together in a practical way.

I would also like to acknowledge the environmental data providers and open-source software communities whose tools and datasets supported the development of the prototype.

12. GLOSSARY

AQI – Air Quality Index

A scale used to communicate air-pollution conditions.

ARIMAX – Autoregressive Integrated Moving Average with Exogenous Variables

A time-series forecasting method that combines historical patterns with external variables.

CAMS – Copernicus Atmosphere Monitoring Service

A service that provides atmospheric and air-quality information.

GFS – Global Forecast System

A numerical weather forecasting system.

MAE – Mean Absolute Error

The average absolute difference between predicted and actual values.

PM2.5 – Fine Particulate Matter

Airborne particles with a diameter of 2.5 micrometres or smaller.

RMSE – Root Mean Squared Error

A forecasting metric that gives more weight to larger prediction errors.

sMAPE – Symmetric Mean Absolute Percentage Error

A percentage-based forecasting error measure.

XGBoost – Extreme Gradient Boosting

A machine-learning technique based on decision-tree ensembles.

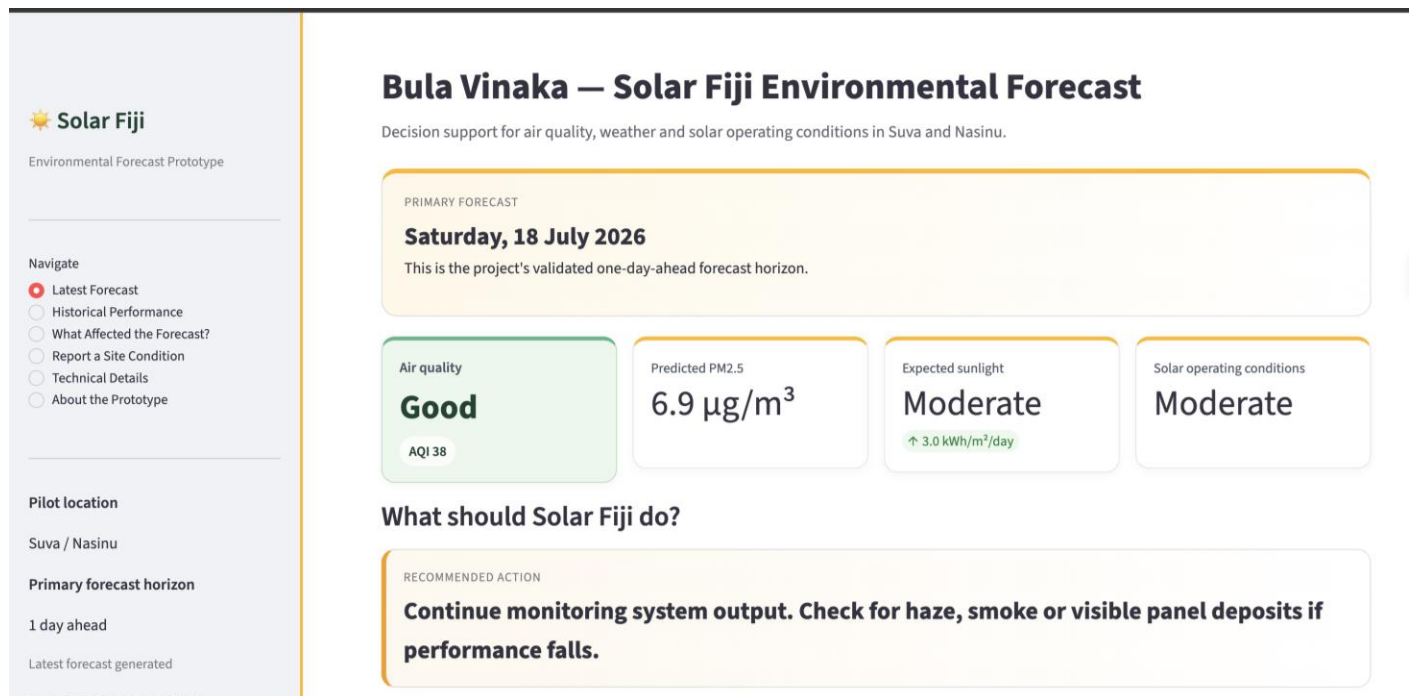
Persistence Baseline

A simple forecasting approach that assumes the next value will be similar to the most recent observed value.

13. APPENDICES

Appendix A – Final Dashboard Screenshots

Figure A1. Latest Solar Fiji Environmental Forecast



Expected environmental conditions

Temperature

21.3 °C

Rainfall

Some rainfall

↑ 1.5 mm

Wind speed

2.8 m/s

External CAMS comparison

6.6 µg/m³

Forecast confidence

The model predicts 6.9 µg/m³, with a 95% forecast interval of 2.9–11.0 µg/m³. The forecast interval crosses the Good–Moderate air-quality boundary, so conditions should be monitored.

The regression model performed best overall but did not detect Moderate AQI days reliably in the historical test period. Forecasts close to an AQI boundary should therefore be interpreted alongside the external CAMS outlook and local site observations.

Figure A2. Seven-Day Environmental Outlook

7-day environmental outlook

Date	Day	Model PM2.5	AQI	Air quality	External CAMS PM2.5	Temperature °C	Rain mm	Wind m/s	Sunlight kWh/m²/day	Operating conditions
18 Jul 2026	1	6.9	38	Good	6.6	21.3	1.5	2.8	3	Moderate
19 Jul 2026	2	7.4	41	Good	8.3	21.6	0.9	2.7	2.6	Moderate
20 Jul 2026	3	6.1	34	Good	9.2	21.6	2.3	1.9	5	Low
21 Jul 2026	4	5.5	31	Good	9	21.4	16.5	1.2	3.5	Low
22 Jul 2026	5	6.2	34	Good	7.5	20.6	3.4	1.9	3	Moderate
23 Jul 2026	6	6.2	34	Good	8.8	20.3	1.5	1.9	3.6	Moderate
24 Jul 2026	7	5.7	31	Good	None	19.2	0.6	1	3.2	Moderate

Day 1 is the project's primary validated forecast horizon. Days 2–7 are extended multi-day planning outlooks and carry greater uncertainty.

Figure A3. Historical Forecast Performance

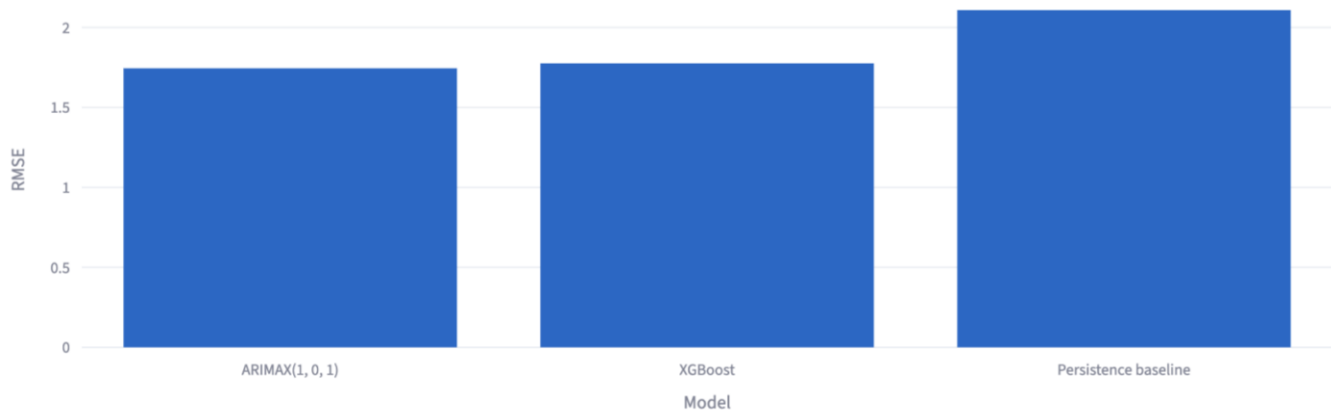
Historical Forecast Performance

The final evaluation used a chronological test period from 25 February 2024 to 10 July 2024.

Model comparison

Model	MAE	RMSE	R2	sMAPE_percent
ARIMAX(1, 0, 1)	1.368	1.745	0.39	27.402
XGBoost	1.411	1.776	0.368	28.45
Persistence baseline	1.54	2.108	0.109	31.329

Test RMSE by Forecasting Model



ARIMAX(1,0,1) achieved the strongest overall regression performance, with RMSE $1.745 \mu\text{g}/\text{m}^3$ and a 17.25% improvement over the persistence baseline.

Figure A4. Actual and Predicted PM2.5

Actual versus predicted PM2.5

Historical Test-Period Forecasts

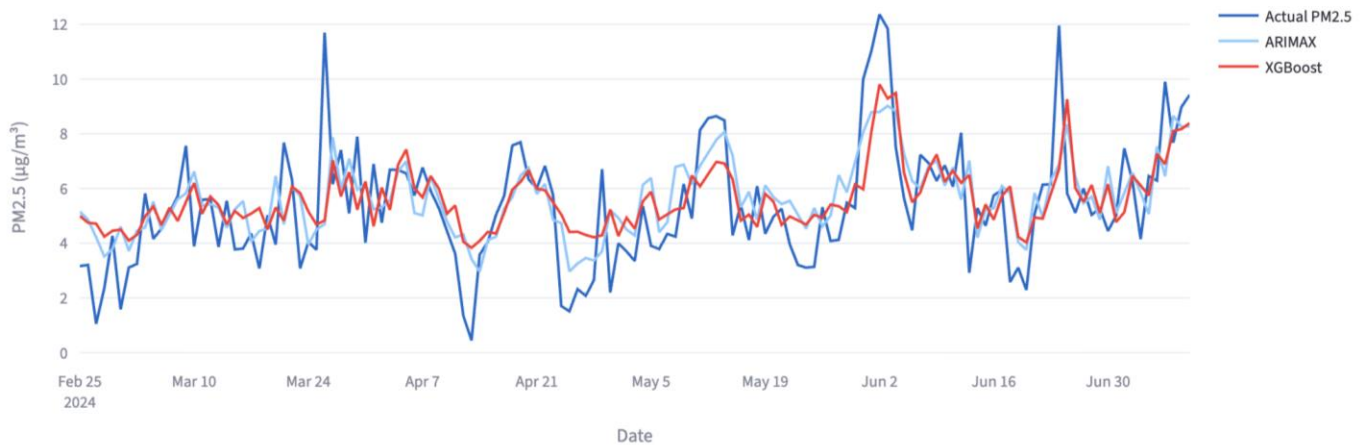
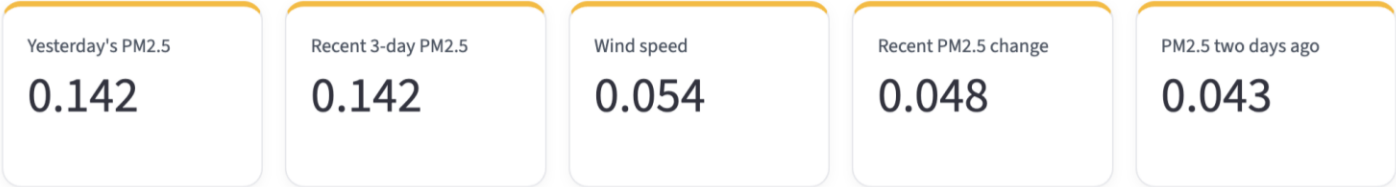


Figure A5. Forecast Drivers

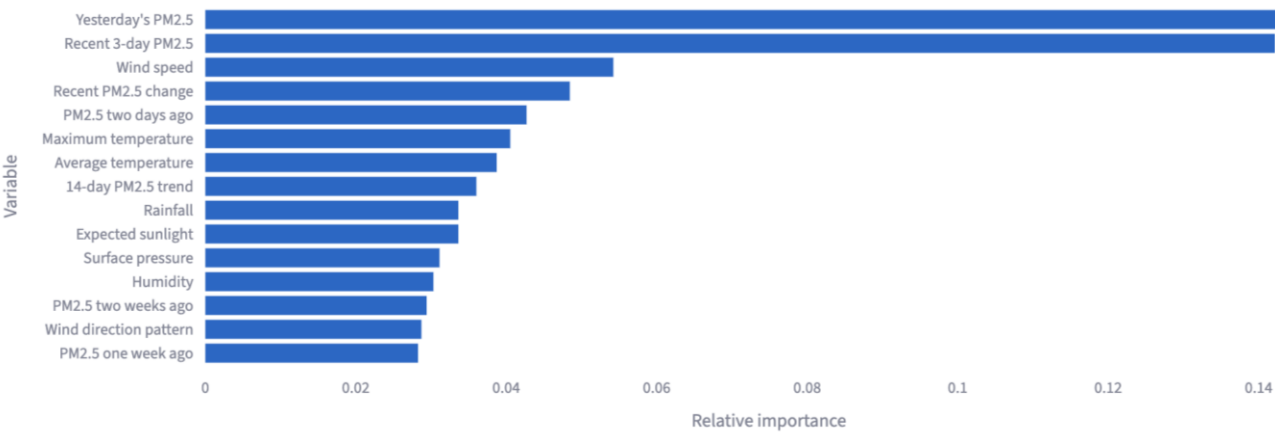
What Affected the Forecast?

XGBoost is retained as an explanatory companion model. Its feature importance helps show which historical and environmental variables contributed most strongly to prediction.

Most influential forecast variables



Top 15 XGBoost Forecast Drivers



The strongest drivers were recent PM2.5 levels and short-term pollution trends. Wind speed, temperature, rainfall and expected solar radiation also contributed to the model. Feature importance indicates predictive contribution, not necessarily direct causation.

Figure A6. Local Site Condition Reporting



Solar Fiji

Environmental Forecast Prototype

Navigate

☐ Latest Forecast

☐ Historical Performance

☐ What Affected the Forecast?

☒ Report a Site Condition

☐ Technical Details

☐ About the Prototype

Pilot location

Suva / Nasinu

Primary forecast horizon

1 day ahead

Latest forecast generated

18 Jul 2026, 10:54 AM Fiji time

Reload latest files

Report a Solar Site Condition

Local observations can provide context that satellite, reanalysis and forecast-model data may not capture.

Site or region

Suva / Nasinu

Site name, if different

Observation date

2026/07/17

Observed conditions

Choose options

Was reduced solar output observed?

Unknown

Additional notes

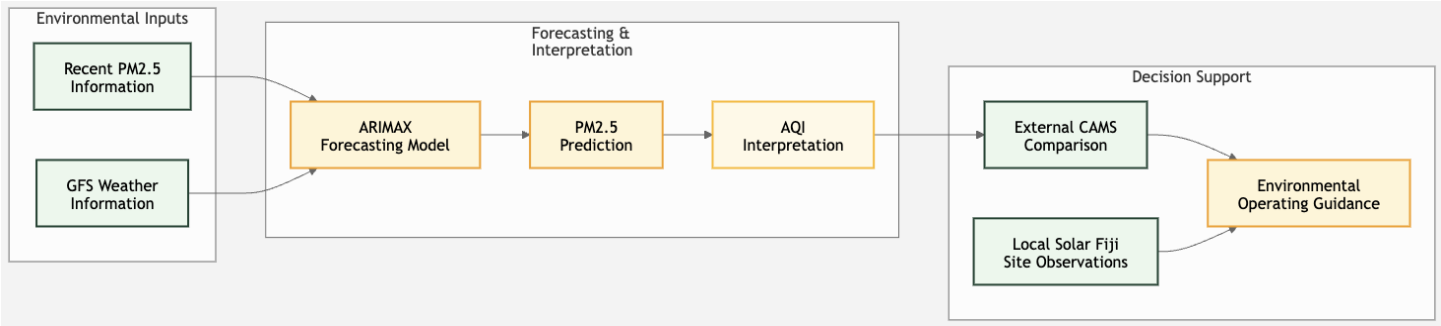
Example: visible haze in the morning, dust on panels, output lower than expected...

Save site report

Appendix B – Final Model Comparison

Model	MAE	RMSE	R ²	sMAPE
ARIMAX(1,0,1)	1.368	1.745	0.390	27.40%
XGBoost	1.411	1.776	0.368	28.45%
Persistence	1.540	2.108	0.109	31.33%

Appendix C – System Workflow



Appendix D – Future System Development

A possible future version of the platform could combine:

- **Local air-quality sensors**
- **Solar inverter and generation data**
- **Weather forecasts**
- **CAMS atmospheric information**
- **Site-specific environmental and solar decision support**

Appendix E – Github Repository

<https://github.com/yukti18/solar-fiji-aqi-forecasting>

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